

Nagaraj Gireppa Kanni

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Software Engineer and AI/ML researcher completing an MS in Software Engineering at SJSU, with hands-on research across OS internals, AV safety validation, and quantum machine learning. Builds full-stack AI systems end-to-end — an LLM agent with swappable model backends, a C++ safety-gate CLI sustaining **15M+ records/sec**. Brings rigorous experimental methodology and a focus on shipping measurable, defensible results.

EDUCATION

San Jose State University

Master of Science in Software Engineering

San Jose, CA

Jan 2024 – Dec 2025

- **Publication: Nagaraj G K et al.**, *Quantum Machine Learning for Image Classification*, Zenodo 2025 — benchmarked hybrid quantum-classical models vs. **CNN baselines** across 3 datasets.

Nitte Meenakshi Institute of Technology

Bachelor of Engineering in Computer Science

Bangalore, India

Sep 2018 – Jul 2022

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C, C++, PHP

Frameworks & Libraries: FastAPI, React, SQLAlchemy, NumPy, Pandas

Machine Learning: PyTorch, TensorFlow, LLM / Agentic AI (Gemini, Claude, Ollama APIs), Model Evaluation, PennyLane, Qiskit

Systems, Infra & DevOps: C++, CUDA, OpenMP, Multithreading, CMake, CTest, Docker, GitHub Actions, Git, pytest

Databases & APIs: MySQL, MongoDB, Schema Design, REST, WebSockets, OAuth, Tableau

EXPERIENCE

San Jose State University

Researcher — AI-Integrated OS Project

San Jose, CA

Jun 2026 – Present

- Unified three independently-built codebases into one working operating system by finding and fixing **12 breaking conflicts**, reviewing the OS's low-level C/x86 code line-by-line against its design specs.
- Fixed an operating-system bug that silently broke task scheduling after **49.7 days** of continuous uptime, by tracing it to an internal counter overflow (a **32-bit tick-wraparound fault**) and applying the fix.
- Reduced manual code review by building a standalone **Python** tool that automatically scans code for duplication, dead code, and conflicts between modules, catching **100%** of bugs in its own test suite.

San Jose State University

Research Volunteer — AV Safety Validation Systems

San Jose, CA

Jan 2025 – Dec 2025

- Prevented safety regressions in self-driving-car sound-detection software before deployment by building an automated test system that generates **12+** realistic driving scenarios and blocks any update that makes detection worse.
- Benchmarked a **TensorFlow/PyTorch** sound-detection model, catching **100% of critical safety sounds** (sirens, glass-break) with **zero false alarms**, despite a modest 27.5% average accuracy across all sound types.
- Delivered live detection alerts to **100+** simultaneous clients, in under **500 milliseconds** each, by building a **FastAPI/WebSocket** streaming backend.

Unisys

Student Researcher — Applied ML

Bangalore, India

Nov 2021 – Apr 2023

- Designed hybrid **quantum-classical** neural architectures for image classification, validated on real **IBM Quantum** hardware and simulators.
- Improved classification accuracy **5%** over CNN baselines by systematically tuning hyperparameters and searching architectures across 9 circuit configurations.
- Uncovered hardware-specific **degradation patterns** by benchmarking model execution across IBM Quantum hardware vs. simulators, informing the published evaluation methodology.

Additional Experience

- **CXanywhere LLC**, Software Engineer (May–Jun 2023) — Automated **Python/SQL** data pipelines for user-behavior analysis, cutting manual reporting **20%**.
- **The Sparks Foundation**, SWE Intern (Jun–Jul 2021) — Shipped a full-stack **banking app** with secure auth and a normalized **MySQL** schema for money-transfer workflows.

PROJECTS

Orbit — AI-Native Personal Finance App | Python, FastAPI, React, LLM Agents

Jan – Apr 2026

- Eliminated manual transaction categorization by building an **LLM agent** that autonomously extracts and tags transactions from statements (PDF/CSV/image) and answers spending questions via **semantic chat**.
- Enabled a privacy-first, zero-telemetry local mode by engineering **hot-swappable LLM backends** (Gemini, Claude, or offline Ollama) atop an async **FastAPI/SQLAlchemy** backend.

Audio-Based ML Framework for Smart Vehicles | YAMNet, PANNs, PyTorch, TensorFlow — MS Capstone, 4-person team

Jan – Dec 2025

- Enabled real-time detection of safety-critical vehicle sounds (sirens, collisions, gunshots) by owning the **edge ML inference pipeline** for a 4-person MS capstone platform.
- Reached sub-500ms end-to-end detection latency across **500+ sound classes** by integrating a dual-model ensemble (**YAMNet** + **PANNS**) via score fusion and tuning audio preprocessing.

ScenarioGate — **AV Telemetry Safety-Gate CLI** | *C++17, CMake, Multithreading*

Apr – Jul 2026

- Automated safety testing for self-driving-car simulations by building a dependency-free **C++17 CLI** that scores each scenario's detection accuracy and false-alarm rate against configurable pass/fail thresholds, ready to plug into a CI pipeline.
- Processed telemetry logs at **15.56M records/sec** — nearly 2x faster using just 2 CPU cores — by designing a custom **thread pool** that splits ingestion work in parallel.

CUDA Tiled Matrix Multiplication | *CUDA, C++, CMake, CTest*

Jun – Aug 2026

- Built a fast **CUDA** matrix-multiplication kernel (32x32 tiled, shared-memory) that also runs on regular CPUs via an OpenMP fallback, with a **CMake** build that auto-detects GPU availability.
- Caught GPU calculation errors automatically by building a **CTest** suite that checks GPU output against a trusted CPU result on every code change, wired into **GitHub Actions** CI.